NON POWER NUCLEAR APPLICATIONS

Desalination

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- Potable water is in short supply in many parts of the world. Lack of it is set to become a constraint on development in some areas.
- Nuclear energy is already being used for desalination, and has the potential for much greater use.
- Nuclear desalination is generally very cost-competitive with using fossil fuels. "Only nuclear reactors are capable of delivering the copious quantities of energy required for large-scale desalination projects" in the future (IAEA 2015).
- As well as desalination of brackish or sea water, treatment of urban waste water is increasingly undertaken.

It is estimated that one-fifth of the world's population does not have access to safe drinking water, and that this proportion will increase due to population growth relative to water resources. The worst-affected areas are the arid and semiarid regions of Asia and North Africa. A UNESCO report in 2002 said that the freshwater shortfall worldwide was then running at some 230 billion m^3/yr and would rise to 2000 billion m^3/yr by 2025. Wars over access to water, not simply energy and mineral resources, are conceivable.

A World Economic Forum report in January 2015 highlighted the problem and said that shortage of fresh water may be the main global threat in the next decade.

Privacy - Terms

Fresh water is a major priority in sustainable development. Where it cannot be obtained from streams and aquifers, desalination of seawater, mineralised groundwater or urban waste water is required. A study in 2006 by the UN's International Atomic Energy Agency (IAEA) showed that 2.3 billion people lived in water-stressed areas, 1.7 billion of them having access to less than 1000 m^3 of potable water per year. With population growth, these figures will increase substantially.

Water can be stored, while electricity at utility scale cannot. This suggests two synergies with base-load power generation for electrically-driven desalination: undertaking it mainly in off-peak times of the day and week, and load-shedding in unusually high peak times.

World Energy Outlook 2016 reported that in 2015, there were about 19,000 desalination plants worldwide, to provide water to both municipal and industrial users. Almost half of global installed desalination capacity was in the Middle East, followed by the European Union with 13%, the USA with 9%, and North Africa with 8%. Globally, seawater is the most common feedwater type, supplying about 60% of installed capacity, followed by brackish water at over 20%.

WEO 2016 also reported on energy consumption for desalination. The UAE used 556 TJ/yr, followed by Saudi Arabia 168 TJ/yr, Qatar 118 TJ/yr, and Kuwait 76 TJ/yr.

Cumulative investment in desalination plants reached about $21.4 billion in 2015 and is expected at least to double by 2020 according to a 2016 report by market analyst, Research and Markets. The report, *Seawater and Brackish Water Desalination*, includes a prediction that investment by 2020 should top $48 billion showing a compound annual growth rate of 17.6%. The report assesses the market for large industrial or municipal facilities with a capacity greater than 1000 m^3/day (m^3/d). It highlights a growing gap between freshwater resources and demand from all sectors.

Desalination

Most desalination today uses fossil fuels, and thus contributes to increased levels of greenhouse gases. Total world capacity in 2016 was 88.6 million m^3/d (32,300 GL/yr) of potable water, in almost 19,000 plants. Of this, 73% is membrane desalination, and 27% thermal, though in the year to mid-2016, 93% of new capacity contracted was membrane. In 2015, over 65% of global installed

desalination capacity was RO. A majority of the plants is in the Middle East and north Africa. Combining power generation and water production by desalination is economically advantageous and is widely used in the Middle East.

In December 2015 the "Global Clean Water Desalination Alliance – H2O minus CO2" initiative was launched at the COP 21 climate talks in Paris, and called on its 17-nation membership to use clean energy to power new desalination plants. The call was part of the alliance's aim to tackle the water-energy nexus and climate change.

The largest desalination plant – the $3.8 billion Al-Jubail 2 in Saudi Arabia – has 948,000 m^3/d (346 GL/yr) MED-TVC capacity, plus 2745 MWe power generation using gas turbines. The Saudi Saline Water Conversion Corporation (SWCC) takes about 62% of output to supply Riyadh. China is building a 1 million m^3/d RO plant to supply Beijing. Two-thirds of the world capacity is processing seawater, and one-third uses brackish artesian water.

Desalination technologies

The two major types of desalination technologies used around the world can be broadly classified as either thermal processes, in which feedwater is boiled and the vapour condensed as pure water (distillate), or membrane desalination processes, in which feedwater is pumped through semi-permeable membranes to filter out the dissolved solids. The main thermal processes are multi-stage flash (MSF) distillation, multi-effect distillation (MED) and vapour compression variants – thermal and mechanical (TVC, MVC). The main membrane process is reverse osmosis (RO). Singapore is investing in electro-deionisation (EDI) as a low-energy option.

More than three-quarters of the capacity is MSF and RO, but MED is increasing rapidly, according to the International Desalination Association.

The major technology in use and being built today is **reverse osmosis (RO)** driven by electric pumps which pressurise water and force it through a semi-permeable membrane against its osmotic pressure*. This accounted for 65% of 2016 world capacity, up from only 10% in 1999. With brackish water, RO is much more cost-effective, though MSF gives purer water than RO. RO relies on electricity to drive the actual process and requires clean (filtered) feedwater.

* IAEA 2015 states that operating pressure for osmosis ranges from 17 to 27 bars for brackish water and from 55 to 82 bars (5500 to 8200 kPa) for seawater. The energy efficiency of seawater RO heavily depends on recovering the energy from the pressurized reject brine. In large plants, the reject brine pressure energy is recovered by a turbine; commonly a Peloton wheel turbine recovering 20% to 40% of the consumed energy.

Hybrid thermal-membrane plants have a more flexible power-to-water ratio, efficient operation even with significant seasonal and daily fluctuations of the electricity and water demand, less primary energy consumption and an increase of plant efficiency, thus improving economics and reducing environmental impacts. MSF+RO or MED-TVC+RO hybrid plants exploit the best features of each technology for different quality products or a blended product.

Several thermal distillation processes capable of using waste heat from power generation are in use: **multi-stage flash (MSF) distillation** process using steam, was earlier prominent. It works by flashing a portion of the water into steam in multiple stages of what are essentially countercurrent heat exchangers and it accounted for 23% of world capacity in 2012. It is more energy-intensive than MED, but it can cope with suspended solids and any degree of salinity. The Japan Atomic Energy Agency (JAEA) has designed a 600 MWt HTR called the GTHTR300 which produces 300 MWe and uses the waste heat in MSF desalination, the projected water cost being half that of using gas-fired CCGT.

An increasing number of plants use **multi-effect distillation (MED)** with 8% world capacity in 2012, or **multi-effect vapour compression (MVC or VCD) distillation** or a combination of these, e.g. MED-TVC with thermal vapour compression. Multiple-effect distillation (MED) is the low temperature thermal process of obtaining fresh water by recovering the vapour of boiling seawater in a sequence of vessels (called effects), each maintained at a lower temperature than the last. Because the boiling point of water decreases as pressure decreases, the vapour boiled off in one vessel can be used to heat the next one, and only the first one (at the highest pressure) requires an external source of heat, such as that from the condenser circuit of a power plant. It is higher-cost than RO but can cope with any degree of salinity. For Kuwait, MED was selected because no pre-treatment of feedwater was required, in an area with algal blooms and organic matter.

Membrane distillation (MD) is an emerging process which is thermally-driven.

Electro-deionisation (EDI) uses an electric field to pull dissolved salts from water. Singapore has plans to scale up the technology and demonstrate it at a 3,800 m^3/d facility at Tuas. Energy consumption of 1.65 kWh/m^3 has been demonstrated at a 50 m^3/d pilot plant.

Forward osmosis (FO) may be used in conjunction with a subsequent process for desalination. The FO draws water through a membrane from a feed solution into a more concentrated draw solution, which is then desalinated without the problems of fouling, such as often encountered with simple RO. FO plants operate in Gibraltar and Oman.

Desalination is energy-intensive. RO needs up to 6 kWh of electricity per cubic metre of water (depending on both process and its original salt content), though the latest RO plants such as in Perth, Western Australia, and Singapore use 3.5 kWh/m^3, or 4 kWh/m^3 including pumping for distribution. Hence 1 MWe continuous will produce about 4000 to 6000 m^3 per day from seawater. Nuclear-powered water desalination with RO has a carbon footprint of about 50 gCO_2eq/m^3, compared with 1700 g/m^3 for gas and 2900 g/m^3 for coal[a].

MSF and MED require heat at 70-130°C and use about 38 kWh/m^3 thermal input, plus 3.5 kWh/m^3 electrical for MSF and 1.5 kWh/m^3 for MED-TVC. (IAEA 2015 quotes 100 kWh/m^3 thermal input, plus 3.5 kWh/m^3 electrical for MSF and 50 kWh/m^3 thermal input, plus 2.5 kWh/m^3 electrical for MED.) A variety of low-temperature and waste heat sources may be used, including solar energy (especially for MED), so the above kilowatt-hour figures are not properly comparable. For brackish water and reclamation of municipal wastewater RO requires only about 1 kWh/m^3. The choice of process generally depends on the relative economic values of fresh water and particular fuels, and whether cogeneration is a possibility. Thermal processes are more capital-intensive.

Desalination dependence

About three-quarters of Israel's water is desalinated, and one large RO plant provides water at 58 cents per cubic metre, claimed to be the world's cheapest. Until 2013 it also claimed to have the world's largest seawater RO plant at Soreq, producing 627,000 m^3/d. Desalinated and reclaimed water is used for agriculture in the Negrev desert. In 2015 Israel and Jordan signed a $900 million agreement for a new desalination plant at Aqaba on the Red Sea, supported by the World Bank and based on a 2013 agreement.

The new agreement involves desalination of 80 million m^3 per year/220,000 m^3/d at the Aqaba plant, with Israel buying half of that amount for use in its southern port town of Eilat and the Arava region – both desert areas with a chronic water shortage. Jordan will get half the water for the arid southern part of that country. As part of the deal, Israel will supply an additional 50 million m^3 of water for the central and northern parts of Jordan from its Lake Kinneret. In addition to the desalination, over 100 million m^3 of concentrated brine will be pumped 180 km north to replenish the Dead Sea.

Malta gets two-thirds of its potable water from RO, and this takes 4% of its electricity supply.

At the end of 2016 desalination met 25% of Singapore's water demand, as one of the island state's Four National Taps, along with local catchment water, imported water, and NEWater, Singapore's own recycled wastewater. In 2016, 55% of water was imported from Malaysia. A further 228,000 m^3/d plant was due online in 2016, supplying potable water at US 22¢/m^3 (compared with US 49 cents and 36 cents for the first and second plants). Singapore wants to increase the proportion of water it gets from desalination and wastewater reuse from 45% today, to 85% by 2060, by which time, industrial use is expected to account for 70% of water demand. Its 137,000 m^3/d Marina East desalination plant being built by Keppel is designed to treat seawater and reservoir water. The raw water intake from both sources goes through a dual flow chamber, with pre-treatment using flocculation and dissolved air flotation, then ultrafiltration. This is followed by a two-pass RO system, and post-treatment using ultraviolet disinfection.

Saudi Arabia in 2011 obtained 3.3 million m^3/d from 27 government-owned (SWCC) seawater desalination plants, 70% of the country's requirements. Twelve plants, accounting for most of production, use MSF and 7 plants use MED, in both cases the plants are integrated with power plants (cogeneration plants), using steam from the power generation as a source of energy for desalination. Eight plants are single-purpose plants that use RO technology and power from the grid. The UAE is heavily dependent on seawater desalination, much of it with cogeneration plants. Algeria in mid-2013 had 2.1 million m^3/d capacity and another 400,000 m^3/d is envisaged.

In February 2012 China's State Council announced that it aimed to have 2.2 to 2.6 million m^3/d seawater desalination capacity operating by 2015, and early in 2015 the target under the government's Special Plan for Seawater Utilisation was 4 million

m^3/d. However, a 2017 report from the State Oceanic Administration said that the total capacity in 2016 was 1.18 million m^3/d. The cost ranged from CNY 5 to 8/m^3 ($0.74 to 1.18/$m^3$). Two-thirds was used for industrial purposes. Some 400 of the 668 largest cities in China are reported to experience water scarcity.

The Kwinana desalination plant near Perth, Western Australia, has been running since early 2007 and produces about 140,000 m^3/d (45 GL/yr) of potable water, requiring 24 MWe of power for this, hence 576,000 kWh/day, or 4.1 kWh/m^3 overall, and about 3.7 kWh/m^3 across the membranes. The plant has pre-treatment, then 12 seawater RO trains with capacity of 160,000 m^3/d which feed six secondary trains producing 144,000 m^3/d of water with 50 mg/L total dissolved solids. The cost is estimated at A$ 1.20/m^3. Discharge flow is about 7% salt. Future WA desalination plants will have more sophisticated pre-treatment to increase efficiency. In August 2011 the state government decided to double the size of its new Southern Water Desal Plant at Binningup plant near Perth to 100 GL/yr, taking the cost to about $1.45 billion. Stage 1 of 50 GL/yr was within the A$ 955 million budget.

Nuclear desalination studies

Concern has sometimes been expressed regarding the environmental effects of concentrated brine discharge to the sea from coastal desalination plants. In 2018 the results of a six-year study in Australia were released, showing that the effects of such brine outfalls were minimal.

Small and medium sized nuclear reactors are suitable for desalination, often with cogeneration of electricity using low-pressure steam from the turbine and hot seawater feed from the final cooling system. The main opportunities for nuclear plants have been identified as the 80-100,000 m^3/d and 200-500,000 m^3/d ranges. US Navy nuclear powered aircraft carriers reportedly desalinate 1500 m^3/d each for use onboard.

Desalination can provide a way to vary substantially the amount of electricity supplied to the grid by a plant operating continuously at full power, in response to varying demand. Surplus power is fed to a RO desalination plant when it is available. The potable water can be stored much more readily than electricity.

A 2006 IAEA report based on country case studies showed that costs would be in the range 50-94 US cents/m^3 for RO, 60-96 ¢/m^3

for MED and $1.18 to 1.48/m^3 for MSF processes, with marked economies of scale. These figures are consistent with later reports. Nuclear power was very competitive at 2006 gas and oil prices. A French study for Tunisia compared four nuclear power options with combined cycle gas turbine and found that nuclear desalination costs were about half those of the gas plant for MED technology and about one-third less for RO. With all energy sources, desalination costs with RO were lower than MED costs.

At the April 2010 Global Water Summit in Paris, the prospect of desalination plants being co-located with nuclear power plants was supported by leading international water experts.

As seawater desalination technologies are rapidly evolving and more countries are opting for dual-purpose integrated power plants (*i.e.* cogeneration), the need for advanced technologies suitable for coupling to nuclear power plants and leading to more efficient and economic nuclear desalination systems is obvious. The IAEA Coordinated Research Program (CRP) New Technologies for Seawater Desalination using Nuclear Energy was organized in the framework of a Technical Working Group on Nuclear Desalination that was established in 2008. The CRP ran over 2009-2011 to review innovative technologies for seawater desalination which could be coupled to main types of existing nuclear power plant. The CRP focused on low temperature horizontal tube MED, heat recovery systems using heat pipe based heat exchangers, and zero brine discharge systems.

An IAEA preliminary feasibility study on nuclear desalination in Algeria was published in 2015, for Skikda on the Mediterranean coast, using cogeneration. The nuclear energy option was very competitive compared with fossil fuels.

Desalination: nuclear experience and plans

The feasibility of integrated nuclear desalination plants has been proven with over 150 reactor-years of experience, chiefly in Kazakhstan, India and Japan. Large-scale deployment of nuclear desalination on a commercial basis will depend primarily on economic factors. Indicative costs are 70-90 US cents per cubic metre, much the same as fossil-fuelled plants in the same areas.

One obvious strategy is to use power reactors which run at full capacity, but with all the electricity applied to meeting grid load

when that is high and part of it to drive pumps for RO desalination when the grid demand is low.

The BN-350 fast reactor at Aktau, in Kazakhstan, successfully supplied up to 135 MWe of electric power while producing 80,000 m^3/d of potable water over some 27 years to 1999, about 60% of its power being used for heat and desalination. The plant was designed as 1000 MWt but never operated at more than 750 MWt, but it established the feasibility and reliability of such cogeneration plants. (In fact, oil/gas boilers were used in conjunction with it, and total desalination capacity through ten MED units was 120,000 m^3/d.)

In Japan, some ten desalination facilities linked to pressurised water reactors operating for electricity production yield some 14,000 m^3/d of potable water, and over 100 reactor-years of experience have accrued. MSF was initially employed, but MED and RO have been found to be more efficient there. South Korea has some MED plants associated with PWRs. The water is used for the reactors' own cooling systems.

India has been engaged in desalination research since the 1970s. In 2002 a demonstration plant coupled to twin 170 MWe nuclear power reactors (PHWR) was set up at the Madras Atomic Power Station, Kalpakkam, in southeast India. This hybrid Nuclear Desalination Demonstration Project (NDDP) comprises a RO unit with 1800 m^3/d capacity and a MSF plant unit of 4500 m^3/d costing about 25% more, plus a recently-added barge-mounted RO unit. This is the largest nuclear desalination plant based on hybrid MSF-RO technology using low-pressure steam and seawater from a nuclear power station. They incur a 4 MWe loss in power from the plant.

In 2009 a 10,200 m^3/d MVC plant was constructed at Kudankulam to supply fresh water for the new nuclear plant. It has four stages in each of four streams. An RO plant there supplied the plant's township initially. The full MVC plant was commissioned in mid-2012, with quoted capacity of 7200 m^3/d to supply the plant's primary and secondary coolant and the local town. The cost is quoted at INR 0.05 per litre (USD 0.9/m^3).

Since 2010 Russia's Rostov nuclear power plant at Volgodonsk has produced make-up water using eight MED units, now totaling 9600 m^3/d. Other plants also use MED and RO for desalination.

Egypt's Nuclear Power Plant Authority plans a two-unit AES-2006 nuclear power plant with desalination facility at El-Dabaa, on the Mediterranean coast, 290 km west of Cairo. Atomstroyexport

quotes the El Dabaa reactors as 3200 MWt, 1190 MWe gross for power generation only, using warm seawater for cooling. However, with desalination (MED + RO) taking 432 MWt from the secondary circuit, they would be 1050 MWe gross, 927 MWe net and each produce 170,000 m^3/d at a cost of less than $1/$m^3$.

In August 2017 a cooperation agreement between Saudi Technology Development Corporation and China Nuclear Engineering & Construction Group (CNEC) set up a partnership project for "developing desalination projects using high-temperature gas-cooled nuclear reactors."

A low temperature (LTE) nuclear desalination plant uses waste heat from the nuclear research reactor at Trombay has operated since about 2004 to supply make-up water in the reactor.

Pakistan in 2010 commissioned a 4800 m^3/d MED desalination plant, coupled to the Karachi Nuclear Power Plant (KANUPP-1, a 125 MWe PHWR) near Karachi, though in 2014 it was quoted as 1600 m^3/d. It has been operating a 454 m^3/d RO plant for its own use.

China General Nuclear Power (CGN) has commissioned a 10,080 m^3/d seawater desalination plant using waste heat to provide cooling water at its new Hongyanhe project at Dalian in the northeast Liaoning province.

Much relevant experience comes from nuclear plants in Russia, Eastern Europe and Canada where district heating is a by-product.

Large-scale deployment of nuclear desalination on a commercial basis will depend primarily on economic factors. The IAEA is fostering research and collaboration on the issue. In the 1960s the US Atomic Energy Commission investigated using nuclear plants up to 10,000 MWt for desalination on the west coast.

In 2014 Rusatom Overseas said it was planning to promote thermal desalination plants using nuclear power on a BOO (build-own-operate) basis. The first meeting of the Rusatom Overseas International Expert Council on Desalination took place in September 2014 in Moscow.

In California, a county's Drought Task Force is teaming up with Diablo Canyon nuclear plant owner Pacific Gas and Electric with a view to using the site's 5700 m^3/d RO desalination plant to supply up to 3100 m^3/d surplus water for county residents.

In South Africa, due to acute water shortages in the region, Eskom announced in May 2017 that it would install a small groundwater

desalination plant at its Koeberg nuclear power plant. It produces water solely for the plant, but Eskom is collaborating with Cape Town authorities on a seawater desalination plant, which would produce 2,500 to 5,000 m^3/d as a demonstration plant for a larger project.

Small nuclear reactors suitable for desalination

SMART: South Korea has developed a small nuclear reactor design for cogeneration of electricity and potable water. The 330 MWt SMART reactor (an integral PWR) has a long design life and needs refuelling only every three years. The main concept has the SMART reactor coupled to four MED units, each with thermal-vapour compressor (MED-TVC) and producing total 40,000 m^3/d, with 90 MWe.

CAREM: Argentina has designed an integral 100 MWt PWR suitable for cogeneration or desalination alone, and a prototype in being built next to Atucha. A larger version is envisaged, which may be built in Saudi Arabia.

NHR-200: China's INET has developed this, based on a 5 MW pilot plant.

Floating nuclear power plant (FNPP) from Russia, with two KLT-40S reactors derived from Russian icebreakers, or other designs for desalination. (If primarily for desalination the twin KLT-40 set-up is known as APVS-80.) ATETs-80 is a twin-reactor cogeneration unit using KLT-40 and may be floating or land-based, producing 85 MWe plus 120,000 m^3/d of potable water. The small ABV-6 reactor is 38 MW thermal, and a pair mounted on a 97-metre barge is known as Volnolom floating NPP, producing 12 MWe plus 40,000 m^3/d of potable water by RO. A larger concept has two VBER-300 reactors in the central pontoon of a 170 m long barge, with ancillary equipment on two side pontoons, the whole vessel being 49,000 dwt. The plant is designed to be overhauled every 20 years and have a service life of 60 years. Another design, PAES-150, has a single VBER-300 unit on a 25,000 dwt catamaran barge.

See also: Small Nuclear Power Reactors paper.

Wastewater and groundwater treatment

for irrigation

In the Middle East, a major requirement is for irrigation water for crops and landscapes. This need not be potable quality, but must be treated and with reasonably low dissolved solids.

In Oman, the 76,000 m^3/d first stage of a submerged membrane bioreactor (SMBR) desalination plant was opened in 2011. Eventual plant capacity will be 220,000 m^3/d. This is a low-cost wastewater treatment plant using both physical and biological processes and which produces effluent of high-enough quality for some domestic uses or reinjection into aquifers.

In Australia AGL plans to install a 2000 m^3/d RO desal plant to treat water from fracking in its Gloucester coal seam gas project. This will be used for irrigation, rather than being potable quality. Also in Australia the Commonwealth Scientific and Industrial Research Organisation (CSIRO) has found that the addition of nutrients could make desalinated water more financially attractive to farmers, who normally pay 20 cents/kilolitre for irrigation water, whereas most desal groundwater costs more than A$1/kL.

New desalination projects

Algeria has undertaken a study on nuclear power generation and desalination using RO and MED. The 500,000 m^3/d Magtaa seawater RO desal plant at Oran costing $495 million was commissioned in November 2014, following the 120,000 m^3/d Fouka seawater RO desal plant at Tipaza near Algiers in 2011, costing $185 million. The country is also considering MSF desalination for two new plants in addition to the 91,000 m^3/d Arzew MSF plant now operating. Total capacity is 2.3 million m^3/d.

Argentina: A 3000 m^3/d seawater RO plant is being built at Puerto Deseado, Santa Cruz, about 1800 km south of Buenos Aries.

Australia: Six major seawater RO plants were commissioned at a cost of A$ 12 billion between 2006 and 2012. However, the Kurnell plant near Sydney is not used but costs some A$500,000 per day on care and maintenance.

In Victoria, a 440,000 m^3/d (150 GL/yr) RO desalination plant near Wonthaggi built by Degremont was commissioned in 2012 to supply Melbourne. It claims to use 90 to 120 MWe of renewable energy, and is expandable to 200 GL/yr. However, it has never been used since 2012 completion and remained on standby to

2017. Melbourne Water's A$ 18 billion in repayments due over 27 years from 2015 is proposed to be spread over 60 years. The A$ 607 million pa maintenance cost is billed to customers. The state government in 2016 ordered 50 GL of water costing A$ 27 million (over the maintenance figure) by mid-2017, and then a further 15 GL before the plant returned to standby.

Adelaide's 100,000 m^3/d (36 GL/yr) plant started operation in 2011, with plans to expand it to 100 GL/yr. A 200-280,000 m^3/d desalination plant to serve the expanded Olympic Dam mine in South Australia has environmental approval but now may not proceed.

Perth has two RO seawater desal plants, a 123,000 m^3/d (45 GL/yr) one (costing A$ 387 million) completed in 2006 powered by a wind farm, and a 100 GL/yr one powered by 65 MWe of dedicated renewable energy, which together provide half the city's needs. Following extensive trials, the city plans a groundwater replenishment scheme from treated wastewater which is expected to be half the cost and use half the energy of seawater desalination. It will include a new Advanced Water Recycling Plant and provide 7 GL/yr from 2016 and 28 GL/yr eventually about 2022.

Brazil: In Ceara state the water utility Cagece is planning to build a 86,400 m^3/d seawater desalination plant to service the capital, Fortaleza. Suez has an agreement for water reuse for the state.

Chile: The main focus is on the Antofagasta region in the north, and the Atacama region immediately south of it. Both extend from the coast inland to the border. The Atacama Desert is in both regions.

Several mining projects in the high-altitude Atacama desert of northern Chile rely on seawater desalination at the coast to supply their water. In November 2015 the Chilean state copper commission Cochilco said that desalination will provide half of the water demand for the country's copper mines by 2026 – 924,000 m^3/d. There are 16 mining-related desalination projects worth US$10 billion planned or under construction in the country, and nine are already operating.

BHP Billiton and Rio Tinto have built a $3.43 billion, 220,000 m^3/d (79 GL/yr) RO seawater desalination plant with twin 1.07 m diameter pipes and pumps for their Escondida copper mine in the Atacama Desert in the Antofagasta region. Seawater is pumped 170 km inland to a reservoir near the mine, 3200 m above sea level and 185 km inland. It requires over 1000 MWe from the grid for

desalination alone and was commissioned early in 2017. Doosan built the plant at Caleta Coloso under Bechtel supervision.

Escondida water desalination project, Antofagasta, Chile (image: Black & Veatch)

BHP's 2017 commitment to a $2.5 billion expansion of its Spence copper mine 1710 m above sea level involves an $800 million, 86,400 m^3/d desalination plant at Mejillones, 60 km north of Antofagasta. It will be built by Caitan, a 50:50 joint venture of Mitsui and Tedagua. Saipam, through its Cobra Montajes subsidiary, will build the 155 km x 900 mm pipeline and three pumping stations. The project is expected to be operational by mid-2020.

Chilean state copper company Codelco is seeking a BOOT (build-own-operate-transfer) agreement to build a 54,000 m^3/d plant expandable to 145,000 m^3/d for its Chuquicamata, Radomiro Tomic, Ministro Hales, and Gabriela Mistra mines – inland in the Antofagasta region. The $1.2 billion project includes a 160 km pipeline and pumping system.

A 104,000 m^3/d seawater RO plant is planned to serve the districts of Caldera, Chañaral, Copiapó, and Tierra Amarilla, in the northern half of the Atacama region, costing $250 million. State-owned utility Econssa awarded an initial contract to a consortium of Spain's GS Inima and Claro Vicuna Valenzuela of Chile in October 2017, with planned operation of the first phase in 2020.

Energias y Aguas del Pacifico (ENAPAC) is a solar-powered 227,000 m^3/d seawater RO desalination plant and water transport project which aims to support expansion of several mining

operations in the Atacama region. Five pumping stations on twin 1.4 m diameter pies will take the water 72 km to a reservoir at Copiapo, 700 m above sea level. The RO process itself will require 3.5 kWh/m^3, including pre and post treatment, and about the same again to deliver the water inland. The whole project will be powered about 80% by a 100 MWe solar PV plant, as well as off-peak grid power. The $500 million Trends project was approved by the country's Environmental Evaluation Commission in September 2018.

China built 112 seawater desalination plants in 2014, with total output of 927,000 m^3/d, according to the State Oceanic Administration.

It is looking at the feasibility of a nuclear seawater desalination plant in the Yantai area of Shandong Peninsula, producing 80-160,000 m^3/d by MED process, using a 200 MWt NHR-200 reactor. A 100,000 m^3/d seawater RO plant supplied by Abengoa of Spain started operating early in 2013 at Qingdao in Shandong province. Another project is for a 330,000 m^3/d plant near Daya Bay.

A 50,000 m^3/d Aqualyng plant was completed in October 2011 at Caofeidian on Bohai Bay in Hebei province, and a second stage doubled this in 2012. The Hong Kong based Beijing Enterprises Water Group (BEWG) with Aqualyng is building a 1 million m^3/d RO plant at Caofeidian for CNY 7 billion to supply Beijing through a 270 km pipeline by 2019, and a 3 million m^3/d plant is planned to expand this to supply the capital, providing about one-third of its needs. The pipeline, itself a major part of the project, will cost about CNY 10 billion, and supply desalinated water at CNY 8/m^3 ($1.28/$m^3$).

In March 2013 the National Development and Reform Commission announced new plans for seawater desalination, including for the cities of Shenzhen and Zhoushan, Luxixiang Island in Zhejiang Province, Binhai New Area in Tianjin, Bohai New Area in Hebei, and several industrial parks and companies. The cost is likely to be some CNY 21 billion ($3.35 billion). China aimed to produce 2.2 million m^3/d of desalinated water by 2015, more than three times the 2011 level, but only reached 1.18 million in 2016. More than half of the freshwater channelled to islands and more than 15% of water delivered to coastal factories would come from the sea by 2015, according to the plan.

The Luo Fang Wastewater Treatment Plant in Shenzhen is to be equipped with membrane bioreactor technology from GE Water &

Process Technologies, adding 150,000 m^3/d and taking capacity to 400,000 m^3/d.

A 300,000 m^3/d seawater desal plant at Tianjin is under construction and will be the first zero-liquid discharge (ZLD) plant in the world. It is due to supply petrochemical plants from 2017.

Cyprus's Water Development Department is calling for bids to build a 15,000 m^3/d RO desalination plant in Kouklia, Paphos, on a build-own-operate-transfer (BOOT) basis. This will augment existing desalination capacity of 220,000 m^3/d. The country aims to produce 30% of water from desalination by 2020, while the goal for reuse water is 20% by 2020, and 30% by 2025, mainly for irrigation.

Egypt's Nuclear Power Plant Authority plans a two-unit AES-2006 nuclear power plant with desalination facility at El-Dabaa, on the Mediterranean coast, 290 km west of Cairo. See section above.

The former largest desalination plant, 24,000 m^3/d RO, at Marsa Matrouh in the northwest has pressure exchanger (PX) energy-recovery devices by Energy Recovery Inc of California.

In 2016 FCC Aqualia won a contract to build a 150,000 m^3/d desalination plant at El Alamein. In 2017 the Engineering Authority of the Armed Forces (EAAF) of Egypt announced plans for three 150,000 m^3/d capacity facilities in Al-Alamain, Al-Jamila, and East Port Said, involving "large French and German companies". Its Ain Sokhna 164,000 m^3/d plant in the Suez Canal Zone is part of an integrated water and power project being built on a BOO basis by Hyflux for delivery in 2018. A small plant is also planned at Najila, Matruh.

FCC Aqualia and Orascom in 2017 won a $320 million contract to build and operate the 1.6 million m^3/d Abu Rawash wastewater reuse plant for Cairo, to produce potable water. It is financed by the African Development Bank.

A $15.6 billion project including four desalination plants on the Sinai Peninsula was announced in April 2018.

In February 2019 it was reported that in addition to the 58 desal plants then operating, total 440,000 m^3/d, 16 new desal plants totalling 671 m^3/d were under construction or otherwise deemed urgent and a further 19 would add 682,000 m^3/d. The total would then be 1.8 million m^3/d.

Ghana: Abengoa has built a 60,000 m^3/d seawater RO plant at Nungua to supply Accra. The $125 million contract covers operation and maintenance for 25 years.

In India, further plants delivering 45,000 m^3 per day are envisaged, using both MSF and RO desalination technology, and building on the extensive experience outlined above. For Chennai, the 100,000 m^3/d Minjur RO seawater desalination plant was commissioned in 2010, the 100,000 m^3/d Nemmeli RO seawater desalination plant was commissioned in 2013, and bids for a 150,000 m^3/d plant there were received in 2017. A 400,000 m^3/d plant is planned at Penur nearby, to open in 2025, also serving Chennai. Also in Tamil Nadu, two 60,000 m^3/d seawater RO plants are being built at Kuthiraimozhi in Ramanathapuram and Alanthalai in the port city of Tuticorin to supply potable water.

In Karnataka four new desalination plants are planned for a satellite city to Bengaluru, comprising a $380 million plant at Mangaluru, a $110 million plant at Udupi, one of $23 million at Kundapura, and at Saligrama, a $12 million plant. IDE Technology and Vagas are likely to build them.

Indonesia: South Korea investigated the feasibility of building a SMART nuclear reactor with cogeneration unit employing MSF desalination technology for Madura Island, and later studies have been on larger-scale PWR cogeneration by Batan.

Iran: A 200,000 m^3/d MSF desalination plant was designed for operation with the Bushehr nuclear power plant in Iran in 1977, but initially lapsed due to prolonged construction delays. It is being completed by AEOI subsequently.

In June 2016 Doosan signed a $186 milllion agreement to build a 200,000 m^3/d seawater RO plant at Bandar Abbas.

Iraq: Basrah has 400,000 m^3/d desalination for saline river water, and a Hitachi-led consortium is building a new 199,000 m^3/d RO plant there, for completion in 2016.

Israel has five desalination plants, at Ashdod, Ashkelon, Eliat, Hadera and the 627,000 m^3/d seawater RO plant at Soreq. It has invited tenders for a 200 GL/yr addition at Soreq to take national capacity to 785 GL/yr, about 85% of demand. It also plans to purchase about 35-40 GL/yr from a planned plant at Aqaba jointly run with Jordan.

Jordan has a 'water deficit' of about 1.4 million m^3 per day and is actively looking at nuclear power to address this, as well as supplying electricity. A small (15,000 m^3/d) seawater RO plant on the Gulf of Aqaba commenced operation in April 2017. In July 2018 Jordan announced plans for a $2.82 billion public-private

partnership development north of Aqaba including a 329,000 m^3/d seawater RO desalination plant, with 55,000 m^3/d first stage.

Kenya has awarded contracts for two desalination plants near Mombasa, total 130,000 m^3/d, to come online in 2021. A further 140,000 m^3/d is envisaged.

Kuwait has been considering cogeneration schemes up to a 1000 MWe reactor coupled to a 140,000 m^3/d desalination plant. Meanwhile Hyundai and a Veolia subsidiary in November 2016 commissioned the $1.7 billion Az-Zour North gas-fired combined cycle 1500 MWe power plant and 486,000 m^3/d MED plant. It accounts for about 10% of Kuwait's power capacity and 20% of its desalination capacity. Stage 2 of the Az-Zour Independent Power and Water Project is out to tender until March 2017. The whole project is 40% privately-owned, largely by Sumitomo and Engie.

Libya: in mid 2007 a memorandum of understanding was signed with France related to building a mid-sized nuclear plant for seawater desalination. Areva TA would supply this. Libya is also considering adapting the Tajoura research reactor for a nuclear desalination demonstration plant with a hybrid MED-RO system.

Mexico has a 21,000 m^3/d El Salitral plant in operation from the end of 2013, and has begun construction of a 22,000 m^3/d plant at San Quintin in Baja California, and another similar one at Ensenada, both using public-private partnerships. A consortium of Spain's FCC Aqualia and Aqualia Mexico won a contract to build and operate a 17,300 m^3/d desalination plant project in Sonora from mid-2018 as a public-private partnership.

A 379,000 m^3/d seawater RO desalination plant is contracted at Rosarito Beach, in Baja California near the US border, with 189,500 m^3/d as phase 1 from 2019, and phase 2 by 2024, which would make it the largest desalination plant in the Western hemisphere. Expected cost including 37 years' operation and maintenance is $490 million, offset by $56 million annual revenue. Ownership would transfer to the state in the late 2050s. It would serve both Mexico and California (Otay Water District). However it is now in doubt due to political and economic changes, and will require a tariff increase to proceed.

Morocco has completed a pre-project study with China, at Tan-Tan on the Atlantic coast, using a 10 MWt heating reactor which produces 8000 m^3/d of potable water by distillation (MED). The government had plans for building an initial nuclear power plant in

2016-17 at Sidi Boulbra, and Atomstroyexport was assisting with feasibility studies for this.

In 2014 Abengoa was awarded a contract to build and run for 20 years a 100,000 m^3/d seawater RO desal plant in Agadir, 45 km from that city. The capital cost is €82 million. This project was then expanded in two stages: first to increase capacity to 150,000 m^3/d; and second to provide 125,000 m^3/d irrigation water. In addition, a further increase to 450,000 m^3/d is envisaged. The value of the project is now €309 million, including €250 million for the actual desalination facilities. The cost of potable water will be about \$0.52/m^3.

Oman relies on large-scale desalination for 76% of its water (20% is from wells). It has had a 128,000 m^3/d plant at Sur since 2009. It commissioned a 45,460 m^3/d seawater RO plant at Barka in November 2013 as a BOO independent water project (IWP), expanding an existing facility to 136,000 m^3/d. Barka 2 will add 120,000 m^3/d. Another BOO project is the 191,000 m^3/d Muscat RO plant with commercial operation from February 2016, replacing the old Ghubrah plant serving the city. The Salalah plant was opened in May 2013 – a 69,000 m^3/d seawater desal plant with 445 MWe gas-fired generation. Another seawater RO plant at Sohar is planned to produce 250,000 m^3/d, on top of 150,000 m^3/d since 2007. Oman Power & Water Procurement Company (OPWP) awarded a BOO contract to Hyflux for a 200,000 m^3/d IWP at Qurayyat, to be operating by May 2017. The 225,000 m^3/d Suwaiq IWP is planned and will bring supplies to over 1.2 million m^3/d. Contracts for Salalah III were expected in 2016, followed by Salalah IV of 100,000 m^3/d. In January 2018 Fisia Italimpianti in a consortium led by ACWA, with Veolia and others, in a joint venture with Abengoa, announced a \$100 million contract for a 113,650 m^3/d RO desalination plant to serve Salalah.

Pakistan: A 10,000 m^3/d RO plant costing US\$ 3 million has been commissioned in the drought-ridden Sindh province, where the government is installing 300 RO plants of about 40 m^3/d.

A 2000 m^3/d plant was commissioned in 2015 at Karwat, for Gwadar city, Balochistan, 700 km west of Karachi. In October 2017 China Pak Investment Corporation acquired the Port City project in Gwadar, a gated community for 500,000 Chinese professionals expected by 2020, and including a 23,000 m^3/d seawater desalination plant. This is part of the China-Pakistan Economic Corridor (CPEC). Another RO plant of 189,000 m^3/d is planned there, 90% funded by China.

Qatar has been considering nuclear power and desalination for its needs, and water demand reached about 1.3 million m^3/d in 2010. The Ras Abu Fontas A2 144,000 m^3/d MSF seawater desalination plant built by Mitsubishi Corporation for $504 million was commissioned in 2015. The $467 million Ras Abu Fontas A3 project – with a 164,000 m^3/d RO plant also built by Mitsubishi is due to operate from March 2017 as the country's first large RO plant. In May 2015 Qatar General Electricity and Water Corporation (QEWC, Kahramaa) selected a Japanese consortium of Mitsubishi Corporation and Tokyo Electric Power Company (Tepco), named K1 Energy, to build an electricity and water plant comprising a 620,000 m^3/d desalination facility and a 2.4 GW gas-fired power station at Umm Al Houl, 20km south of Doha, to come online in 2018.

Russia: A new 10,000 m^3/d seawater RO plant is being built offshore near Vladivostok, for commissioning over 2011-12. It is designed for severe climatic conditions.

Saudi Arabia's Saline Water Conversion Corporation (SWCC) operated 5.1million m^3/d of desalinated water capacity in 2017 and is aiming for 7.3 million m^3/d by 2020. Coupling desalination plants with power generation so as to use reject heat reduces energy requirements for desalination by about half. Hence dual-purpose or hybrid plants are favoured, as independent water and power production (IWPP) facilities. Most are on a build-own-operate (BOO) basis. In 2016 the government said it would sell SWCC's entire portfolio of 29 desalination plants with capacity of 4.6 million m^3/d and 7305 MWe on the east and west coasts. It would then build seven new projects totalling 2.6 million m^3/d through public-private partnerships by 2020. The largest of these are (in million m^3/d): Jubail – 1.17; Rabigh 3 – 0.6; Yanbu – 0.45; and Shuqaiq – 0.38.

Shuaibah/Shoaiba. Capacity of 850,000 m^3/d MSF and 150,000 m^3/d RO is in place here on the Red Sea coast 90 km south of Jeddah. In April 2017 Abengoa was contracted to build a 250,000 m^3/d RO plant at Shoaiba 3 for $257 million on a build-own-operate (BOO) basis. At the same time Doosan was awarded a $422 million EPC contract by SWCC for a 400,000 m^3/d RO plant here. This will make Shuaibah/Shoaiba 3 into the world's largest desalination complex.

Ras Al Khair. The 1,025,000^3/d Ras Al Khair (Ras Azzour) MSF project northwest of Jubail on the Gulf coast, cost SAR 27 billion ($7.2 billion) and was built by Doosan from 2010. The project includes a 2.6 GWe power plant. The hybrid desalination facility has a capacity of 727,000 m^3/d MSF evaporation and 307,000 m^3/d

RO membrane filtration. It will supply water to 3.5 million people in the Riyadh area.

Yanbu. SWCC was expanding its Yanbu desalination plant on the Red Sea to supply the Medina region. Phase 1 is a 146,000 m^3/d hybrid plant, mostly MSF using heat recovered from a gas turbine power plant, but with two RO units. Phase 2 upgraded this and added a 68,000 m^3/d MED plant from Doosan using the heat from an associated 690 MWe power plant, all costing over $1 billion. It became the world's largest MED plant. From 2012 Doosan also built Yanbu 3, a 550,000 m^3/d MSF plant. Samsung Engineering had the $1.4 billion EPC contract for 3100 MWe turbines in connection with this, but that contract was cancelled at the end of 2016. In May 2017 Sepco III Electric Power & Construction from China took over the 60% complete work.

Shuqaiq. This is a 212,000 m^3/d RO plant with 850 MWe power generation on the southern Red Sea coast, due to be more than doubled in size.

Rabigh 3. The 134,000 m^3/d plant with 360 MWe generating capacity and steam production on the Red Sea near Yanbu is being expanded with 600,000 m^3/d RO.

Marafiq/Jubail. Capacity of 800,000 m^3/d with 2743 MWe of generating capacity is in place here. Veolia has a $402 million contact to build a 178,600 m^3/d ultrafiltration and RO plant for Marafiq at the $19.3 billion Sadara petrochemical complex on the Gulf coast, to come online in mid-2015. The water will be for two cooling towers and as boiler feedwater. A further 1.17 million m^3/d capacity is planned here.

Khafji. The first of three phases of the King Abdullah Initiative for Solar Water Desalination was operating by 2014. Phase 1 involved construction of two solar plants to generate 10 MWe of power for a 30,000 m^3/d reverse-osmosis (RO) desalination plant at Al Khafji, near the Kuwait border. Phase 2 involves construction of a 60,000 m^3/d RO desalination plant over three years to 2018 by Abengoa, supplied by 15 MWe of polycrystalline PV. The RO plan will have six trains, allowing best use of variable power input. The third phase aims to implement the solar water initiative throughout Saudi Arabia, with the eventual target of seeing all the country's desalination plants powered by solar energy by 2020. One of the main objectives of this initiative under King Abdullah City for Science & Technology (KACST) is to desalinate seawater at a cost of less than Riyal 1.5/m^3 (US$ 0.40/m^3) compared with the current cost of thermal desalination, which KACST says is in the range Riyal

2.0-5.5/m^3 (US$ 0.53-1.47/m^3), and desalination by RO, which is Riyal 2.5-5.5/m^3 (US$ 0.67-1.47/m^3) for a desalination plant producing 30,000 m^3/d.

Saudi Arabia's General Establishment for Water Desalination (GEWD) is, over the four years to 2019, implementing new projects with a total production capacity of up to 2.5 million m^3/d in the Makkah region and the eastern province.

SWCC is setting up three 50,000 m^3/d floating desalination plants in the Red Sea. These can be moved around to supplement any coastal RO plants as required. They are expected in operation at Shuqaiq for Jazan and Asir provinces about October 2019.

Singapore's water agency PUB in 2005 commissioned a large SingSpring RO seawater desalination plant supplying 136,000 m^3/d – 10% of needs, at 49 US cents per cubic metre, and in 2013 commissioned the 318,500 m^3/d Tuaspring RO plant as the second desalination plant on a build-own-operate (BOO) basis, costing US$ 700 million, to provide water at 36 ¢/m^3. It has run at a loss. Both plants are operated by Hyflux. A third desalination plant, the $217 million Tuas 137,000 m^3/d plant using RO and other membrane technology was built by HSL and opened in mid-2018. It is owned and operated by PUB and draws on both seawater and reclaimed sources (NEWater). The fourth desalination plant will be at Marina East, a 137,000 m^3/d BOO project built by Keppel, to supply water to PUB at a first-year price of SGD 1.08/m^3 from both seawater and reservoirs. The fifth desalination plant will be 137,000 m^3/d, on Jurong Island. PUB also has the 228,000 m^3/d Changi Water Reclamation Plant opened in mid-2009, which uses RO to produce NEWater from sewerage. It is the largest of several NEWater plants on the island and is planned to triple in size. It was built by an 80% subsidiary of Beijing Enterprises Water Group on a BOO basis for 25 years at a first-year price of SGD 0.276/m^3 (US$ 0.22).

South Africa: Veolia is building a 1700 m^3/d seawater desalination plant at Lamberts Bay, Cederberg municipality, upgradable to5000 m^3/d. This will be the seventh plant along the west and south Cape coasts installed by Veolia. A 450,000 m^3/d plant costing $1.23 billion is planned for Koeberg, near Cape Town.

Spain is building 20 RO plants in the southeast to supply over 1% of the country's water. Spain has 40 years of desalination experience in the Canary Islands, where some 1.1 million m^3/d is provided.

Tunisia opened its first desalination plant, of 50,000 m^3/d, on the island of Djerba in May 2018. The water utility SONEDE is looking at the feasibility of a cogeneration (electricity-desalination) plant in the southeast of the country, treating slightly saline groundwater. It plans a tender for a 100,000 m^3/d plant at Sfax and signed a loan agreement for this in July 2017. A further 50,000 m^3/d plant capable of doubling in size by 2027 is planned for Zarat, in the Gabes district, by 2021.

The UAE uses a lot of MSF capacity compared with others. It is planning a 68,000 m^3/d plant at Ras Al Kaimah. Sembcorp is expanding the Fujairah 1 RO plant of 136,000 m^3/d, to bring its UAE capacity to 591,000 m^3/d of which 307,000 m^3/d is RO and 284,000 m^3/d is MSF. Also the Shuweihat S2 IPP and seawater desal plant at Al Ruwais started full operation in 2011 and provides 1510 MWe and 454,000 m^3/d by MSF. The Fujairah 2 plant is hybrid SWRO-MED and produces 454,600 m^3/d. The Taweelah A1 cogeneration plant produces 1430 MWe and 385,000 m^3/d and Umm Al Nar produces 394,000 m^3/d. The 91,000 m^3/d Al Hamriya RO desal plant with 400 MWe power station opened in June 2014 to supply Sharjah near Dubai, as part of a 636,000 m^3/d and 2500 MWe complex. The 136,400 m^3/d Al Zawra seawater desal plant is planned at Ajman. GdF Suez has a 25-year power and water supply agreement with Abu Dhabi for the Mirfa project, including a new 136,380 m^3/d RO plant and 1100 MWe power plant, costing $1.5 billion, alongside three existing 34,095 m^3/d MSF units and a power plant.

The Emirates' Federal Electricity and Water Authority (FEWA) announced in April 2016 that it plans four further desalination facilities at a cost of more than Dh 3 billion ($750 million) to produce 600,000 m^3/d, all on a public-private partnership (PPP) basis. The first RO plant of 205,000 m^3/d and costing $260 million will be in Umm Al Quwain emirate for operation from 2020. The second will be 136,000 m^3/d costing $170 million in Ajman emirate, also to start in 2020 (this may be the same as mentioned above for Al Zawra). The third will be identical, taking capacity at Ghalilah in Ras Al Khaimah emirate from 68,000 to 205,000 m^3/dand to operate from 2026. The fourth was planned as a 180,000 m^3/d, $260 million plant operating from 2023 with location to be determined. In October 2018 FEWA let contracts for three plants of 205,000 m^3/d in the Ras Al Khaimah, Umm Al Quwain and Fujairah emirates.

In Dubai the Jebel Ali M cogeneration plant opened in 2013 with 6x243 MWe gas turbines and 8 MSF units providing 640,000 m^3/d. Policy then shifted to decoupling power production from

desalination, and using solar energy plus waste heat for the latter. In March 2018 the Dubai Electricity and Water Authority (DEWA) awarded a $237 million contract to Acciona Agua and Besix for a 182,000 m^3/d brownfield RO plant at Jebel Ali. The target for RO by 2030 is 1.4 million m^3/d out of a total 3.4 million m^3/d anticipated by then. Earlier, Dubai invited bids for constructing a 450,000 m^3/d seawater desalination plant as part of its Hassyan independent power project, but then announced its deferral.

In the UK, a 150,000 m^3/d RO plant is proposed for the lower Thames estuary, utilising brackish water.

USA-Mexico: The 375,000 m^3/d Rosarito seawater plant in Baja, California, is to supply potable water on both sides of the border. A 22,000 m^3/d seawater desalination plant is contracted for San Quintín, Baja.

USA: San Antonio, Texas, is building a 60,000 m^3/d RO desal plant for brackish water from aquifers, to operate from 2016 and costing $193 million. Additions are planned to take capacity to 150,000 m^3/d by 2026.

A $1 billion, 200,000 m^3/d salt water desal plant at Carlsbad, California, opened in 2015. It will require about 40 MWe in full operation and can provide 10% of the San Diego county's potable water. San Diego has ordered a 415,000 m^3/d RO wastewater treatment plant costing $3.5 billion. It is expected to meet one-third of the city's daily drinking water requirement by 2035, making it the second largest potable reuse plant in the USA.

Most or all these have requested technical assistance from IAEA under its technical cooperation project on nuclear power and desalination. A coordinated IAEA research project initiated in 1998 reviewed reactor designs intended for coupling with desalination systems as well as advanced desalination technologies. This programme, involving more than 20 countries, is expected to enable further cost reductions of nuclear desalination.

Other CO2-free desalination

Renewable energy sources are able to be used for desalination more readily than for most electricity supply, since the product can be stored on any scale, unlike electricity. Also electricity can be borrowed from the grid and repaid when the wind is blowing or the sun shining. A 45 GL/yr RO plant at Perth, Western Australia is powered by electricity ostensibly from a wind farm. A new 100

GL/yr RO plant is powered by 65 MWe of dedicated renewable energy (10 MWe solar PV, 55 MWe wind).

Notes & references

Notes

a. Figures based on 3.5 kWh/m^3 for desalination using reverse osmosis, and median life-cycle emissions figures given in Table A.III.2 of Annex III, *Technology-specific Cost and Performance Parameters* in *Climate Change 2014: Mitigation of Climate Change Working Group III Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. [Back]

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